To overcome the above problems, FET can be developed and become important electronic device in the integrated circuit (IC) technology. The FET is a device in which the flow of current through the conducting region is controlled by an electric field (voltage).

There are two types of field effect transistor,

1. Junction Field Effect Transistor (JFET)
2. Metal Oxide Semiconductor Field Effect Transistor(MOSFET)

6.9.1 Junction Field Effect Transistor (JFET)

JFET is a three terminal semiconductor device in which current conduction is by one type of carrier (i.e.) electrons or holes. It is a unipolar device. It has high input impedance and low noise level.


Jun-ichi Nishizava

History: The FET was first patented by Julius Edgar Lilienfeld in 1926 and by Oscar Heil in 1934. The first type of JFET was the static induction transistor (SIT), invented by Japanese engineer Jun-ichi Nishizava and Y. Watanabe in the year 1950.

There are two types of JFET. They are

1. N-Channel JFET
2. P-Channel JFET

6.9.1.1 N-Channel JFET

Construction: Figure 6.9 shows the construction of n-channel JFET and its symbol. It consists of a uniformly doped N-type semiconductor bar made of silicon. On both sides of this N-type semiconductor bar, two heavily doped P-type regions are formed by diffusion. The two P-regions are internally connected and a single lead is taken out, which is called Gate (*G*). Ohmic contacts are made at the two N-type semi-conductor bar. One lead is called as Source(*S*) and the other as Drain (*D*). These two terminals may be interchanged. The source(*S*) is a terminal through which the majority carriers (electrons in the *N* type bar) enter the bar. The drain (*D*) is a terminal through which the majority carriers leave the bar.

Since the two P-regions are heavily doped and N-type bar is lightly doped, two depletion layers are formed in the N-type bar as shown in Figure 6.9. The region between the two layers is called channel. The majority carriers move from

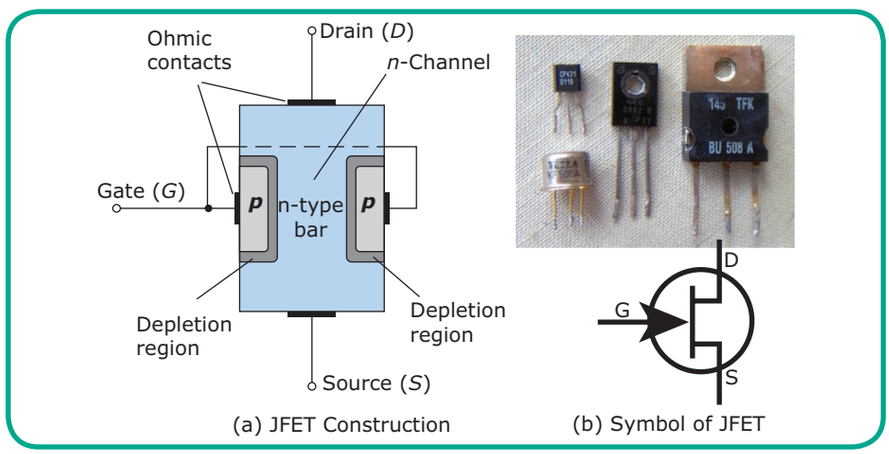


Figure 6.9 Construction of N-Channel JFET and its Symbol

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